

MedAI

System Architecture Document

A comprehensive technical specification of the MedAI platform — covering infrastructure, data flows, AI integration, security, and scalability design.

DOCUMENT

SAD v1.0

BACKEND

Firebase (Auth, Firestore, Storage, FCM)

AI

llmu-glm-5.1 via llmu.ai API

MOBILE

Flutter (Dart) — iOS & Android

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SECTION 1 — SYSTEM OVERVIEW

Architecture Overview

```
graph TD
    subgraph CLIENT_TIER [CLIENT TIER]
        direction TB
        MA[Mobile App iOS + Android]
        RWP[React Web Portal]
        P[Patients - Dart/Flutter Hospital Managers - React]
    end
    subgraph BACKEND_TIER [BACKEND TIER]
        direction TB
        FAuth[Firebase Auth]
        FStorage[Firebase Storage]
        FCloudMsg[Firebase Cloud Messaging]
    end
    subgraph AI_TIER [AI TIER]
        direction TB
        ILMU[OpenAI-compatible - model: llmu-glm-5.1]
    end
    subgraph HOSPITAL_TIER [HOSPITAL TIER]
        direction TB
        HB[Hospital Backend]
    end
    MA --> FAuth
    MA --> FStorage
    MA --> FCloudMsg
    MA --> ILMU
    RWP --> FAuth
    RWP --> FStorage
    RWP --> FCloudMsg
    RWP --> ILMU
    P --> FAuth
    P --> FStorage
    P --> FCloudMsg
    P --> ILMU
    P --> HB
    HB --> FAuth
    HB --> FStorage
    HB --> FCloudMsg
    HB --> ILMU
```

Technology Stack

Layer	Technology	Version/Notes
Mobile UI	Flutter + Dart	SDK ≥ 3.0, Provider state management
Web Portal	React	Frontend-only, no backend required
Authentication	Firebase Auth	Email/password, JWT tokens
Database	Cloud Firestore	v25.1.4, real-time streams
File Storage	Firebase Storage	Avatars, medical documents
Push Notifications	Firebase Cloud Messaging	iOS (APNs) + Android (FCM)
AI / NLP	llmu-glm-5.1 via llmu.ai	OpenAI-compatible API, JSON mode
HTTP Client	Dart http package	v1.2.2
Image/File Picker	image_picker + file_picker	Camera, gallery, PDF
Secure Storage	flutter_secure_storage	API keys, sensitive tokens

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SECTION 2 — COMPONENT BREAKDOWN

Mobile App — Component Architecture

```
lib/
├── main.dart
├── firebase_options.dart
├── constants.dart
├── firestore_serialization.dart
├── appointment_status.dart
├── specialist_model.dart
├── chat_message_model.dart
├── services.dart
├── auth_wrapper.dart
├── storage_service.dart
├── http_calls.dart
├── ai_chat.dart
├── screens.dart
├── ai_chat.dart
├── hover_scale.dart
└── platform_utils.dart
```

AI Chat State Machine

```
ChatState: idle --startConversation()--> collectingSymptoms | AI detects <SYMPTOMS_COMPLETE> | --generatingVerdict--> generateVerdict() [JSON mode] | --bookingPrompt / \ acceptBooking() declineBooking() / \ --collectingBookingPrefs declined | AI detects <BOOKING_COMPLETE> | --searchingSlots--> Firestore slotCache query | --showingSlots--selectSlot()--> booked
```

AI Integration Design

Symptom Collection

- OpenAI-compatible chat completion API
- System prompt instructs collection of 6 fields: mainComplaint, duration, severity (1-10), associatedSymptoms, medicalHistory, medications
- Completion detected via `<SYMPTOMS_COMPLETE>[json]` tag
- Full conversation history passed each call for context

Verdict Generation

- JSON mode request for structured output
- Returns: `specialty`, `urgency`, `explanation`
- Urgency maps to: ASAP / Within a week / Routine
- Fallback to "General Practice" on API failure

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SECTION 3 — SECURITY & SCALABILITY

Security Architecture

Authentication & Authorization

- Firebase Auth issues JWT tokens; all Firestore reads/writes verify `request.auth.uid`
- Users can only read/write their own documents
- Hospital data is read-only for authenticated users
- API keys stored in `flutter_secure_storage`, never in plain `SharedPreferences`
- llmu.ai API key stored server-side only (in app constants, not exposed in JS)

Data Privacy

- Patient passport & phone stored in Firestore (encrypted at rest by Firebase)
- Medical documents uploaded to Firebase Storage with per-user path isolation: `documents/{userId}/...`
- AI chat history stored per-user; no cross-user data access
- Only **free slots** fetched from hospitals — no patient/occupancy data transferred

Production Firestore Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId} {
      allow read, write: if request.auth.uid == userId;
    }
    match /appointments/{id} {
      allow read, write: if request.auth.uid == resource.data.userId;
    }
    match /hospitals/{id} {
      allow read: if request.auth != null;
    }
    match /aiChats/{id} {
      allow read, write: if request.auth.uid == resource.data.userId;
    }
  }
}
```

Transport Security

- All Firebase SDK traffic over TLS 1.3
- Hospital API calls use Bearer token in Authorization header
- Android: `network_security_config` enforces HTTPS
- iOS: App Transport Security (ATS) enforced

Hospital API Security

- Each hospital receives a unique Bearer token from MedAI
- Token validated on every incoming request by hospital backend
- Postback URL protected with shared secret header

Scalability Design

Concern	Design Decision	Scales To
Patient database	Cloud Firestore — serverless, auto-sharded	Millions of users
Slot availability	<code>slotCache</code> : collection with 1-hour TTL; batched hospital fetches	1000s of hospitals
AI calls	Stateless HTTP; conversation history managed client-side	Concurrent users × API rate limit
File storage	Firebase Storage CDN — per-user paths	Petabyte-scale
Push notifications	FCM topics / device tokens via Firestore	Billions of devices
Real-time updates	Firestore onSnapshot() — WebSocket streams	Low-latency at scale

Deployment & CI/CD

Mobile Releases

Flutter build pipelines via GitHub Actions. Android APK → Play Store; iOS IPA → App Store Connect.

Firebase Rules

Firestore & Storage rules versioned in `firestore.rules`. Deployed via `firebase deploy --only firestore:rules`.

Environment Config

Firebase project-per-environment (dev / staging / prod). `firebase_options.dart` generated per env via `FlutterFire CLI`.

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